Advanced Time Series Coursework

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1. Introduction

The fMRI (functional Magnetic Resonance Imaging) dataset contains data about the dynamic activity of each of the 70 brain regions defined in the Desikan atlas for 24 patients, measured thorugh changes in the blood-oxygen-level-dependent (BOLD) signal during resting state fMRI (R-fMRI) sessions. The data were gathered during the NKI1 study (INSERT CITATION), where each patient was simply asked to "stay awake with eyes open" while dynamic activity was being measured. There are one or two time series per patient, each comprising 404 measurements of the BOLD signal taken at lags of 1400 ms from one another.

In fMRI data the BOLD signal can be imagined as a proxy measurement for neural activity (REFERENCE RIC), which makes time series models for the time-varying location well-suited for the task of describing the data generating process. In fact, neural activity can be imagined as a latent signal evolving over time of which we only observe a noisy version, (the BOLD signal). In order to recover the latent signal, we need to apply some type of filtering to the noisy observed values; this is exactly the conceptual setting behind models for the time-varying location.

Figure 1: Posterior cingulate cortex

In this report we will focus on two time series extracted from the fMRI dataset:

1. y_1: the dynamic activity in region 59 for patient 16 during scan 1.
2. y_2: the dynamic activity in region 59 for patient 13 during scan 1.

Region 59 in the Desikan atlas corresponds to the right-hand posterior cingulate cortex, shown in Figure 1, which plays a role in spatial navigation, self-identity, and switching attention between internal and external tasks. The two time series are plotted in Figure 2.

y_1 and y_2 were chosen, respectively, for the (approximately) gaussian, and non-gaussian behavior. In this report we fit two models for the time varying location on each time series: one which assumes gaussian noise in the measurements, and one which assumes t-distributed noise. The former is cast into a state-space framework and estimated using the Kalman Filter, while the latter is cast into the framework of Generalized Autoregressive Score (GAS) models. The objective of this report is to understand how the two models compare to one another, and in particular how they differ in performance when moving to a non-gaussian setting.

In section 2 we give a brief preliminary analysis of the time series. In section 3 we fit the model assuming gaussian errors on both time series, while in section 4 we fit the model assuming t-distributed errors. In section 5 we discuss the differences in the performance of the two models.

2. Preliminary Analysis

The time series $y1$ is plotted in Figure 1. The series appears stationary in mean and variance. The global mean of the series is 0, and the local mean (represented by the green line in the plot), also hovers around 0. The overall variance of the series is 1.69; the skewness is -0.174, and the kurtosis is 2.87, which are quite close to what we would expect from a normal distribution.Figure 3 shows the ACF and PACF for y_1. The acf shows a geometric decay of the spikes; the pacf also decays quickly to 0, only showing significant spikes at lag 1 and (although barely) at lag 2. This behaviour is compatible with an AR(1) or ARMA(1,1) process.

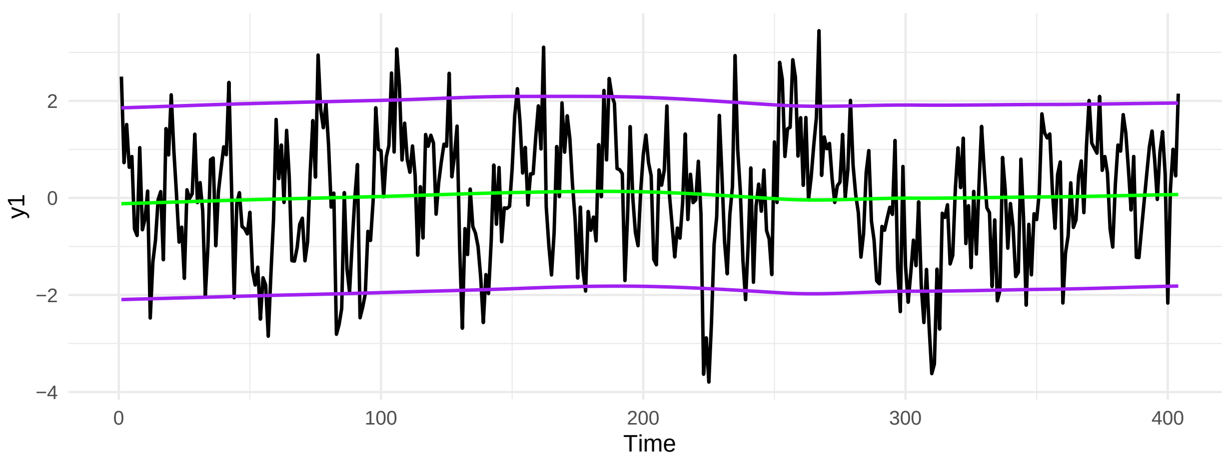

Figure 2: Time series y_2 with robust loess smooth for mean and standard deviation.

3. Time-varying Location model with Gaussian Noise

As mentioned in the introduction, the BOLD signal can be seen as the noisy version of a latent signal, which in our case can be imagined as neural activity. Let y_t denote the measurement at time t, and let μ_t denote the latent signal at time t. The simplest statistical model that can represent the process generating the noisy measurements is the following:

$$y_t = \mu_t + \epsilon_t \ , \ \epsilon_t \overset{iid}{\sim} N(0, \sigma^2_\epsilon) \qquad (1)$$

$$\mu_{t+1} = \omega + \phi\mu_t + \eta_t \ , \ \eta_t \overset{iid}{\sim} N(0, \sigma^2_\eta) \qquad (2)$$

$$\{\epsilon_t\} \perp\!\!\!\perp \{\eta_t\}$$

This model is an example of a state space model. Equation (2) is the "state equation", which describes how the latent signal (or state) evolves over time. The latent signal is what we refer to as the time-varying

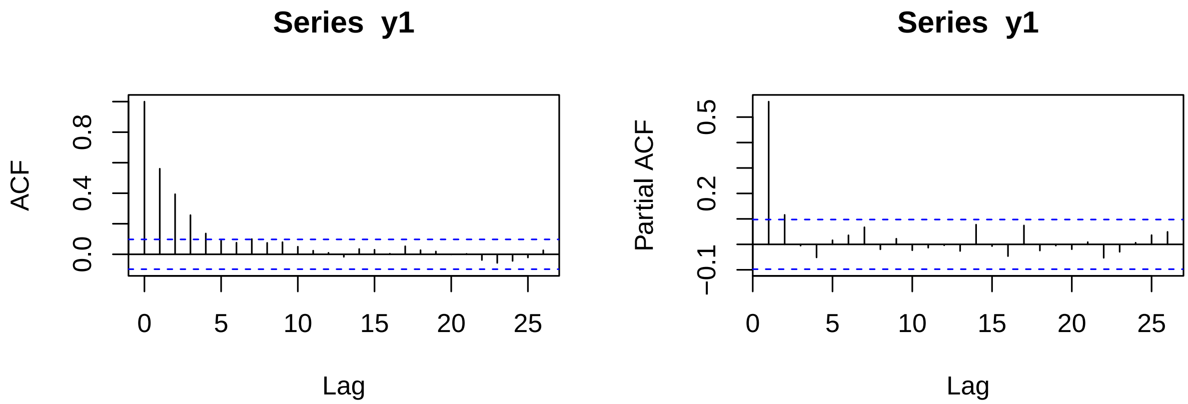

Figure 3: ACF and PACF for y_1

location, since it can be imagined as a location parameter evolving over time. In particular, here we are assuming the evolution of the state is autoregressive of order 1, (i.e. it only depends directly on the previous state), with gaussian IID error terms. Equation (1) is the "measurement equation", which describes how our observed values are only a noisy version of the latent signal. In particular, here we are assuming that the noise added to measurements is gaussian IID, and independent from the noise driving the evolution of the unobserved state. Because of this, the model described above is also referred to as an AR(1) + gaussian noise model. The gaussianity assumption may be unrealistic at times, (as we will see when estimating the model for y_2), and departures from it may harm the accuracy of our model.

Of course, more complex state space models could be formulated, however it seems sensible to start with the simplest one to see how it does. Besides, it is possible to show that the model described above can be reformulated as an ARMA(1,1) model, which makes its use even more sensible based on what we noted in the preliminary analyses of section 2.

The static parameters in the model, ω, ϕ, σ_ϵ, σ_η, can be estimated using Maximum Likelihood (ML) estimation, whereas the latent states are estimated using a recursive algorithm known as the Kalman Filter. One can prove the Kalman Filter provides optimal estimates for the states - in a Minimum Mean Squared Error (MMSE) sense - if the gaussianity assumptions are fulfilled. If these fail, the Kalman Filter can still be applied, but it will no longer be optimal.

We now fit the model to both y_1 and y_2 and comment on the results. For details on the software used for fitting the models, see the Appendix.

3.1 Model fitting

AR(1) signal + gaussian noise model for y_1

```
              Estimate Std.Err z_stat p_value
omega           0.0084   0.043    0.2    0.84
phi             0.6938   0.058   12.0    0.00
sigma_epsilon   0.5609   0.097     NA      NA
sigma_eta       0.8421   0.089     NA      NA


log.likelihood            AIC
          -229            465
```

From our estimates, we see that the estimated autoregressive coefficient ϕ is significantly different from 0, suggesting there is a significant time dependence between successive values of the signal. The intercept ω is not significantly different from 0, which makes sense considering that our series appears stationary around 0. Finally, the signal's random error has an estimated standard deviation which is larger than that of the noise. The state estimates for y_1, obtained through the Kalman Filter are reported in Figure 3.

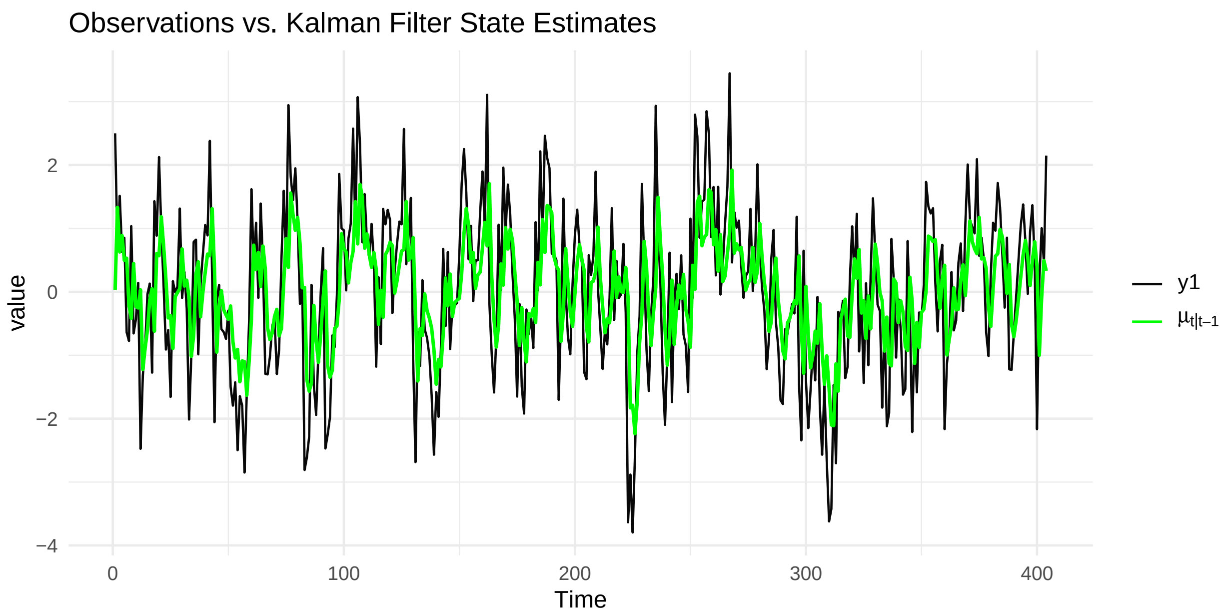

Figure 4: Kalman Filter state estimates for y_1

AR(1) signal + gaussian noise model for y_2

```
              Estimate Std.Err z_stat p_value
omega            0.037  0.0721  0.513   0.608
phi              0.872  0.0315 27.712   0.000
sigma_epsilon    1.407  0.1211     NA      NA
sigma_eta        1.545  0.1423     NA      NA

log.likelihood            AIC
          -539           1086
```

Again, the estimated autoregressive coefficient ϕ is significantly different from 0, suggesting there is a significant time dependence between successive values of the signal. The dependence seems even stronger than for y_1, seeing as the estimated coefficient is lager. As in the model for y_1, the intercept ω is not significantly different from 0. The most striking difference between the 2 models is actually in the estimated standard deviations: in the model fitted to y_2 they are much larger than in the model fitted to y_1. Of course, this makes sense if we consider that we observe a much more variability in the values of y_2 compared to y_1, as noted in section 2. The state estimates for y_2, obtained through the Kalman Filter are reported in Figure 4.

3.2 Model Diagnostics

As mentioned in the beginning of this section, the Kalman Filter relies on some assumptions, and in particular on gaussianity. If these assumptions were to fail, the filter would lose its optimality. We will now run some

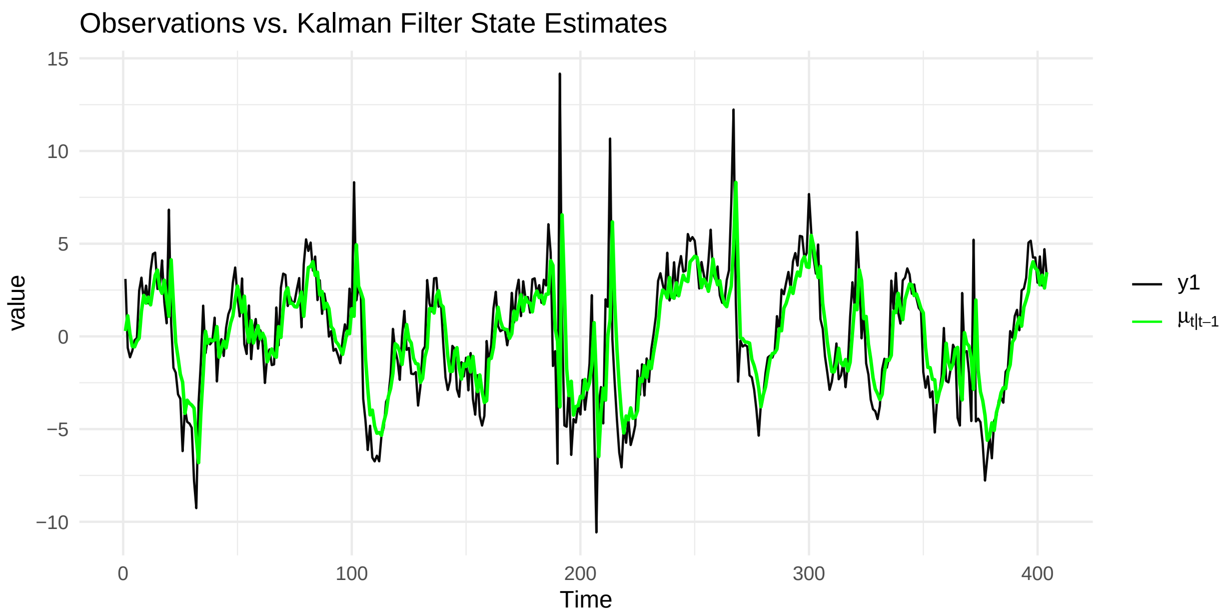

Figure 5: Kalman Filter state estimates for y_2

diagnostics on the two models to see if the assumptions are fulfilled. Diagnostics for the Kalman Filter rely on quantities known as innovations. The innovation at time t, denoted as v_t, is defined as the one-step-ahead prediction error, i.e.:

$$v_t := y_t - \mu_{t|t-1}$$

Where $\mu_{t|t-1}$ denotes the Kalman Filter estimate for the latent μ_t based on the past information set $\mathcal{F}_{t-1}$.

It is possible to show that if the Kalman Filter assumptions are fulfilled, the innovations form a Martingale Difference Sequence (MDS) and are distributed $v_t \overset{iid}{\sim} N(0, F_t)$, Where F_t is the innovation variance in the Kalman Filter. To check if the Kalman Filter assumptions are fulfilled, we must check whether the innovations follow the expected behaviour.

Actually, to simplify things, we will work with the standardized innovations $F_t^{-\frac{1}{2}} v_t$, which are still a MDS and are distributed iid $N(0, 1)$, making it easier to check for and detect deviations from the expected behavior.

Diagnostics for the AR(1) signal + gaussian noise model for y_1 The standardized innovations from the Kalman Filter applied to y_1 and their histogram are shown in Figure 5, their autocorrelation function is shown in Figure 7. A qq-plot for the standardized innovations is shown in figure 9. The standardized innovations seem to hover around 0 and look like a white noise; their acf shows no significant spikes. A Box-Ljung test computed on 20 lags does not reject the hypothesis that the innovations are IID (p-value = 0.7671), further supporting the idea that the model is appropriate. The histogram, the qq-plot, and an estimated kurtosis of 2.95702 suggest that the normality assumption is reasonable; a Shapiro-Wilk test (p-value = 0.8439) and a Jarques-Bera test (p-value = 0.8557) both do not reject the normality assumption. Overall the Kalman Filter assumptions seem appropriate for this model.

Diagnostics for the AR(1) signal + gaussian noise model for y_2 The standardized innovations from the Kalman Filter applied to y_2 are shown in Figure 5, their autocorrelation function is shown in Figure 8. A qq-plot for the standardized innovations is shown in figure 9. Again, the standardized innovations seem to hover around 0 and look like a white noise; their acf shows no significant spikes. A Box-Ljung test computed on 20 lags does not reject the hypothesis that the innovations are IID (p-value = 0.7887). However, the histogram and the qq-plot show some concering behavior: clearly the distribution of the innovations is heavy tailed, in fact it has a kurtosis of 13.60813; a Shapiro-Wilk test (p-value ≈ 0) and a Jarques-Bera test (p-value ≈ 0) both strongly reject the normality assumption. In light of this, the Kalman Filter assumptions do not seem appropriate, which means the filter is non optimal.

The main issue with using the Kalman Filter with heavy-tailed data is that the filter is very sensitive to extreme observations; if an extreme value is observed at time t, the estimated state for time $t+1$, $\mu_{t|t-1}$ will be heavily influenced by that value, which generally reduces the quality of the estimates, because in a sense we are interpreting noise as signal. Score driven models try to tackle this issue by using state updates that are robust to extreme values.

```
    Box-Ljung test

data:  st_v1
X-squared = 11.656, df = 16, p-value = 0.7673


    Shapiro-Wilk normality test

data:  st_v1
W = 0.99764, p-value = 0.8433


    Jarque Bera Test

data:  st_v1
X-squared = 0.3157, df = 2, p-value = 0.854


    Box-Ljung test

data:  st_v2
X-squared = 11.388, df = 16, p-value = 0.785


    Shapiro-Wilk normality test

data:  st_v2
W = 0.89533, p-value = 5.199e-16


    Jarque Bera Test

data:  st_v2
X-squared = 2002, df = 2, p-value < 2.2e-16

[1] 2.956337

[1] 13.65639
```

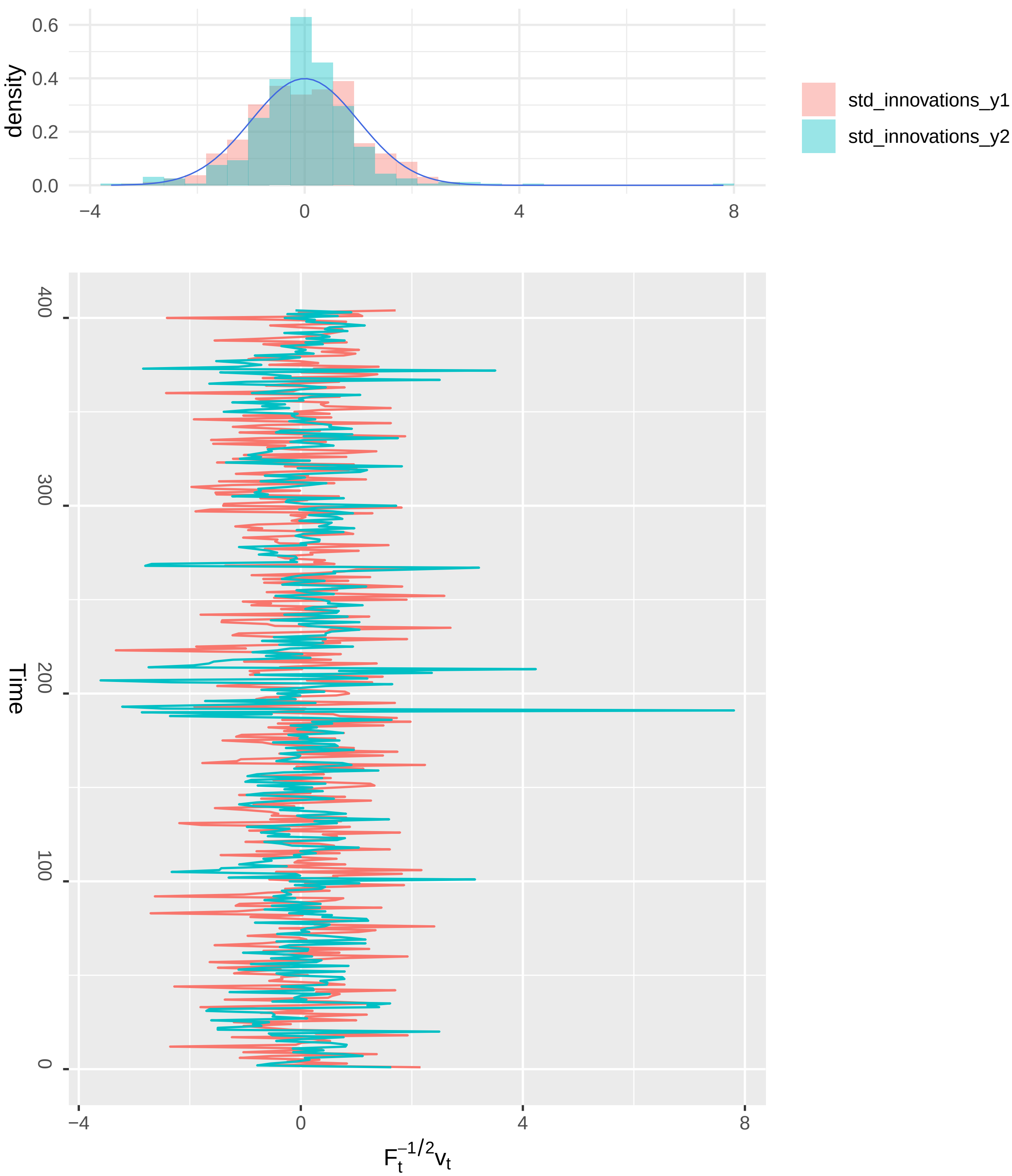

Figure 6: Time series of standardized innovations and Histograms. The blue line in the histogram is a standard normal density.

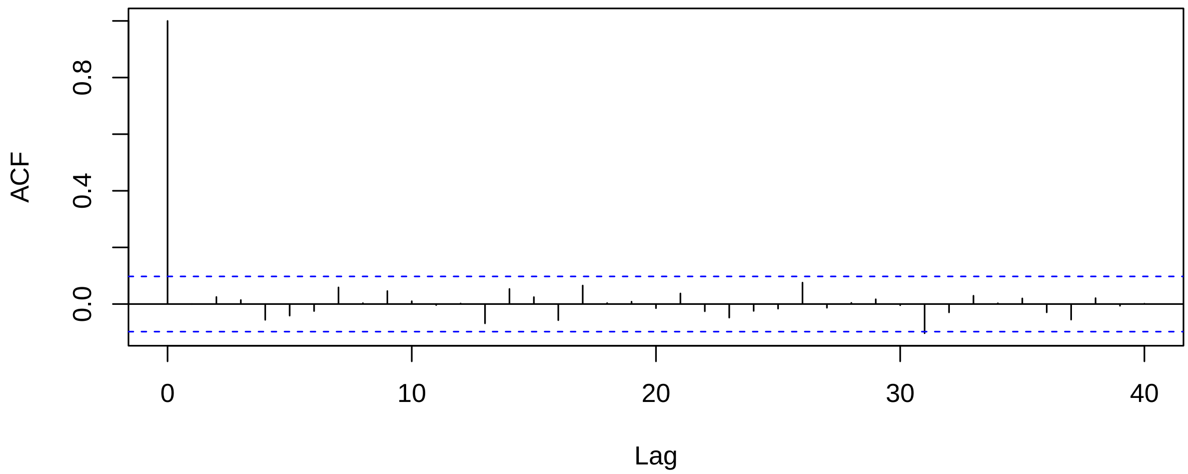

Figure 7: ACF for standardized innovations for y1

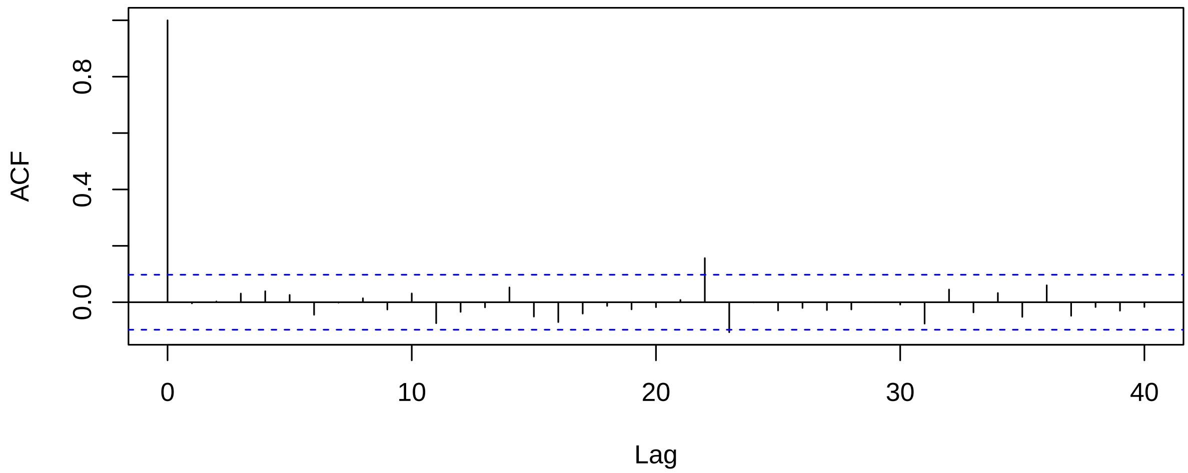

Figure 8: ACF for standardized innovations for y1

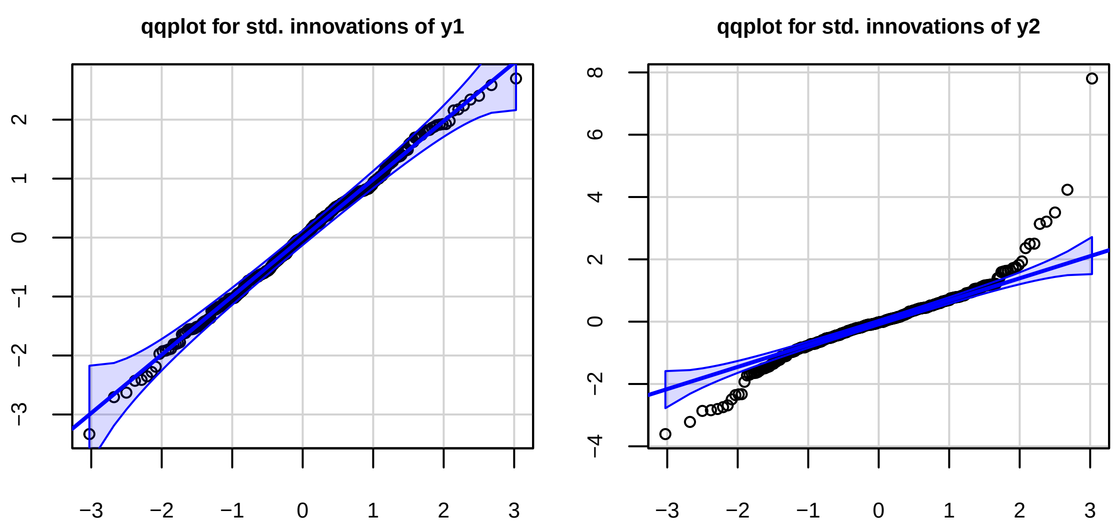

Figure 9: QQplots for standardized innovations